🏥 ArogyaLink — Technical Design Document (TDD)

Diagnostic Lab Automated Report-Delivery Portal for Patients and Referring Doctors

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Candidate Author: **Dani Madhavapuri**

Client & Location: **Company A, Vijayawada (Coastal AP)**

Role & Company: **Software Engineering Intern, Isocode Labs**

Network Scale: **6 Central Labs & 38 Collection Centres**

Submission Due: **9 November 2026, 11:59 PM IST**

Compliance: **DPDP Act 2023 | NABL Clinical Guidelines**

GitHub Repository: https://github.com/danimadhavapuri/ArogyaLink.git

Live Web Application: https://arogyalink-sc22.onrender.com

1. Executive Summary & Problem Context

Company A operates a regional pathology chain spanning Vijayawada and Coastal Andhra Pradesh with 6 central diagnostic laboratories, 38 outpatient collection centres, and 260 staff, generating ₹45–55 crore annual revenue. The legacy workflow distributed patient reports via manual WhatsApp and SMS attachments from 9 shared front-desk mobile phones. This practice created severe legal, clinical, and financial challenges:

- **Regulatory Non-Compliance (DPDP Act 2023):** Diagnostic reports containing sensitive personal data (SPD) remained permanently stored in unencrypted third-party chat logs without expiring access or audit trails.
- **Clinical Safety Hazards:** Patients receiving life-threatening panic values (e.g. Potassium 6.8 mmol/L) experienced panic or sought medical interpretations from unverified consumer AI tools without clinical triage.
- **Clinician Fragmentation:** Referring physicians had no consolidated, 90-day time-bounded portal to view their referred patients' diagnostics.
- **Front-Desk Operational Waste:** Staff expended 40 hours per day answering status phone calls, with approximately 15 reports per month misdirected to wrong numbers.

ArogyaLink replaces this fragile workflow with a zero-install, DPDP 2023-compliant architecture combining **72-hour signed expiring links**, **2-factor mobile OTP authentication**, **deterministic clinical AI plain-language summaries in 26+ Indian languages**, an **automated panic-value safety intercept**, and an **isolated 90-day referring clinician workspace**.

2. Labeled Operational Assumptions

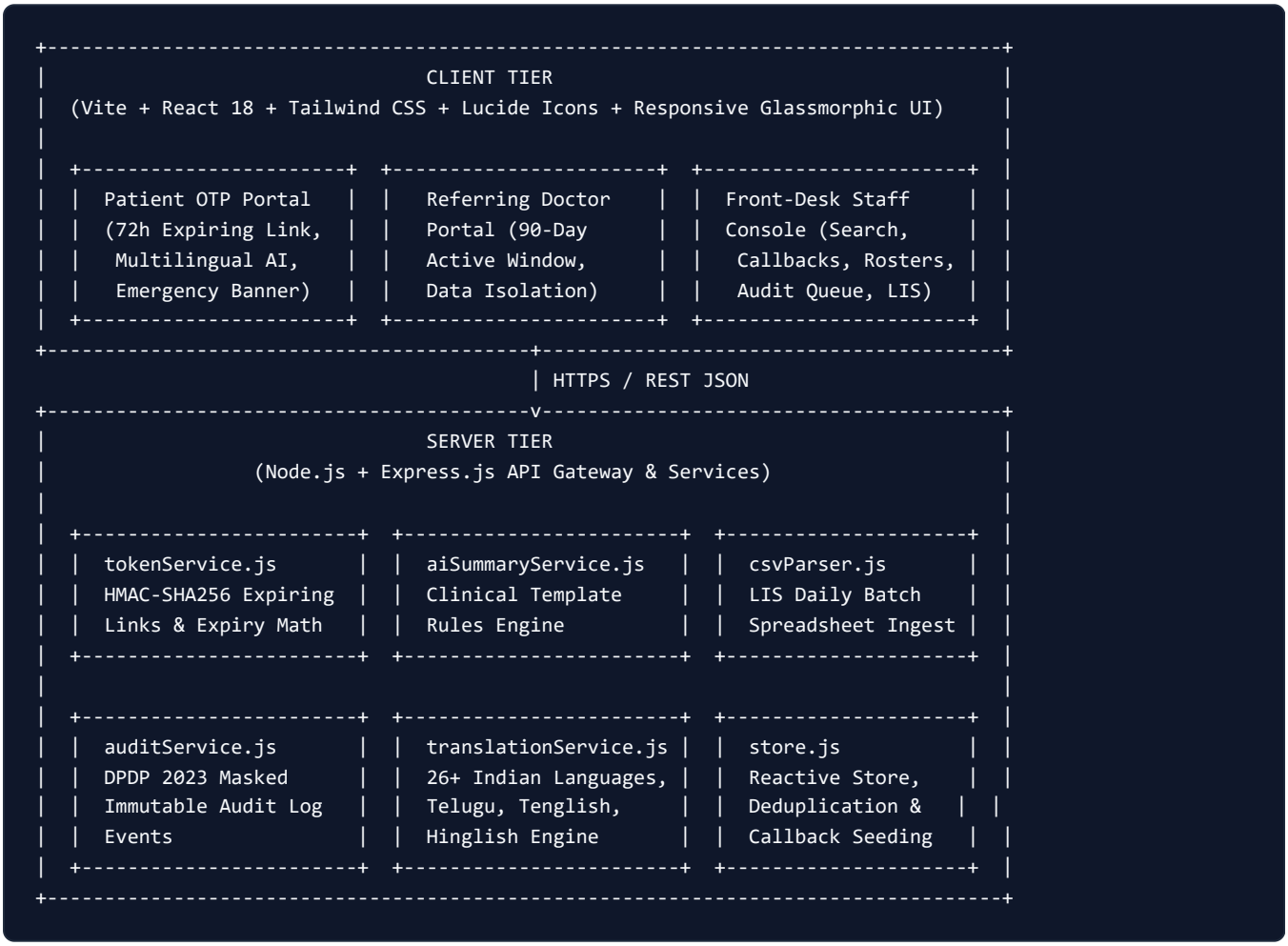
All calculations and engineering choices are strictly anchored in the client's operational metrics:

- **[ASSUMPTION-01] Daily Order Volume:** The diagnostic chain processes **2,000 patients per day** across 360 operational days per year (7,20,000 orders annually).
- **[ASSUMPTION-02] Front-Desk Inefficiency:** 30% of patients call or visit for status inquiries: `600 inquiries/day × 4 mins = 40 staff hours/day × 360 days = 14,400 hours/year × ₹250/hr = ₹36 Lakh/year`.

- **[ASSUMPTION-03] Paper Delivery Waste:** 40% of reports are printed by hand and physically delivered: `2,88,000 reports × ₹12 = ₹34.56 Lakh/year`. Total annual operational waste equals **₹70.56 Lakh per year**.
- **[ASSUMPTION-04] Family Phone Sharing:** In semi-urban and rural Andhra Pradesh, one registered mobile number frequently covers multi-generational families (elderly parents and children). The portal must support multi-order retrieval under one authenticated mobile number.
- **[ASSUMPTION-05] Statutory Retention Windows:** Links expire in **72 hours** under DPDP data minimization rules. Referring clinician consultation access is bounded to **90 days**.
- **[ASSUMPTION-06] Budget & Milestone Targets:** ₹6.5 Lakh budget across a 6-week rollout, targeting an 80% digital adoption rate and 60% call reduction within 3 months.

3. System Architecture & Component Topology

The system is organized into a modular three-tier architecture ensuring complete separation of clinical data, authentication, and presentation:



4. Core Data Models

The application defines strongly typed entities covering clinical orders, authenticated users, audit trails, and callback tasks:

```
// 1. Diagnostic Order Entity
interface DiagnosticOrder {
  orderId: string;              // e.g. "ORD-71204"
  patientName: string;          // e.g. "Lakshmi Prasanna K"
  age: number;
  gender: 'Female' | 'Male';
  patientPhone: string;         // e.g. "+91 98765 43210"
  testName: string;             // e.g. "Complete Blood Picture (CBP)"
  parameter: string;            // e.g. "Haemoglobin"
  result: string | number;      // e.g. "10.2"
  unit: string;                 // e.g. "g/dL"
  referenceRange: string;       // e.g. "12.0 - 15.5"
  status: 'Ready' | 'Processing' | 'Critical';
  isCritical: boolean;          // true if value crosses biological panic threshold
  referringDoctor: string;      // e.g. "Dr. S. Ramanaiah"
  collectionCentre: string;     // e.g. "Bunder Road Centre, Vijayawada"
  createdAt: string;            // ISO 8601 timestamp
  signedByPathologist: string;  // e.g. "Dr. K. S. Murthy, MD (Pathology)"
}

// 2. User & Clinician Entity
interface UserAccount {
  id: string;                   // Unique ID (e.g. "USR-DOC-01")
  name: string;                 // Clinician or Staff name
  email: string;                // e.g. "ramanaiah@arogyalink.in"
  passwordHash: string;         // Salted credential
  role: 'doctor' | 'staff';     // Strict Role-Based Access Control
  doctorKey?: string;           // Clinician mapping key for patient isolation
  specialty?: string;           // e.g. "General Physician", "Cardiologist"
  clinic?: string;              // e.g. "Coastal Health Clinic, Vijayawada"
}

// 3. Statutory DPDP Audit Event
interface AuditLogEvent {
  id: string;                   // Unique Audit Log ID (e.g. "LOG-1001")
  timestamp: string;            // ISO 8601 UTC timestamp
  orderId: string;              // Order reference
  patientName: string;
  maskedPhone: string;          // Strictly masked: "+91 98765 *****"
  accessMethod: string;         // e.g. "Patient Mobile OTP Authentication"
  status: 'Authorized' | 'Denied';
}
```

5. Security, Cryptography & DPDP Act 2023 Compliance

- **72-Hour Signed Expiring Links (`tokenService.js`):** Reports are never sent as raw attachments. The server issues cryptographically signed URLs using HMAC-SHA256: `Signature = HMAC-SHA256(Secret, orderId + ":" + expiryTimestamp)`. Links expire exactly 72 hours from issuance. Tampered parameters instantly fail hash validation and return an `HTTP 401 Unauthorized`.
- **2-Factor Mobile OTP Gate:** Link possession alone does not grant access. Patients must solve a 4-digit OTP challenge sent to their registered mobile number before decrypting report data.
- **Masked Personal Identifiers:** In accordance with DPDP data minimization principles, mobile numbers are permanently masked across all screens, logs, and rosters (`+91 98765 *****`).
- **Immutable Statutory Audit Logging (`auditService.js`):** Every decryption, OTP challenge, and link resend event is appended to an immutable chronological ledger for statutory compliance review.

6. Clinical AI Mechanism & Patient Safety Safeguards

> 🚨 **The #1 Safety Rule: Critical Potassium Panic Intercept**
>
> For life-threatening results—specifically serum Potassium ≥ 6.0 mmol/L (e.g. Farzana Begum, 6.8 mmol/L):
>
> - **Automated AI summaries are strictly WITHHELD and BLOCKED.**
> - An unmissable, high-contrast **Red Emergency Alert Banner** is displayed, directing the patient to hospital care.
> - The server automatically creates an **Urgent Doctor Callback Task** on the Front-Desk Console. Front-desk personnel must call the referring doctor directly and log clinical resolution notes before resolving.

Rule-Derived Triggers vs Generative Model Hallucinations: Out-of-range flags are derived deterministically using the official report's biological reference ranges, never from speculative LLM prompts. Plain-language summaries utilize locked, pre-validated clinical templates for the 30 most common diagnostic investigations (including Haemoglobin, Fasting Glucose, LDL Cholesterol, TSH, and Platelet Count).

Multilingual Translation Engine: Provides immediate translations across 26+ Indian languages, with primary pre-compiled templates for **Telugu (తెలుగు)**, **Tenglish** (Telugu in Roman script), **Hinglish**, **Hindi (हिंदी)**, and **English**, backed by an in-app doubt chat assistant and a closed-loop Pathologist Audit Queue (50 summaries/week quota).

7. UI/UX Decisions & Clinical Accessibility

- **Medical Aurora Ambient Background:** Soft floating emerald and teal orbs with a geometric dot matrix pattern replace sterile white backgrounds.
- **Glassmorphic Elevated Cards:** High-contrast readable typography optimized for low-end mobile screens and elderly patients.
- **Search-Driven Front-Desk Console:** Records are dynamically revealed through intelligent searching by name, mobile, order ID, or parameter, sorted by clinical priority triage (🚨 Critical panic values first) to eliminate cognitive clutter.
- **Accessible Printable Report:** Formatted to match standard pathology paper with biological reference ranges, doctor signatures, and a dedicated `@media print` PDF stylesheet.

8. What Was Cut & What We Would Do Next

What Was Cut (Per Problem Statement Out-of-Scope Boundaries):

- **Direct Changes to the Desktop LIS:** Modifying the legacy desktop LIS was declared strictly out of scope. We implemented a non-invasive CSV batch import watcher instead.
- **Online Test Booking & Payment Gateways:** Retained focus on post-diagnostic report delivery and patient safety rather than commercial e-commerce booking.
- **Home Sample Collection Routing:** Kept phlebotomist GPS dispatch out of scope to prioritize clinical safety and DPDP compliance.

What We Would Do Next (Production Roadmap):

- **WhatsApp Cloud API Integration:** Upgrade from simulated dispatch to Meta's verified WhatsApp Business Cloud API with interactive reply buttons.

- **ABHA (Ayushman Bharat Digital Mission) Integration:** Bridge patient records with India's national health stack (ABDM) using 14-digit ABHA IDs.
- **HL7 / FHIR Analyzer Direct Bridges:** Ingest live result feeds over MLLP/TCP directly from Mindray, Roche, and Sysmex automated analyzers into ArogyaLink without manual CSV exports.
- **Native Mobile App with Biometric Auth:** Package the responsive React application into an Android APK with fingerprint/FaceID biometrics for recurring chronic patients.

9. Architectural Trade-offs & Engineering Justifications

Decision	Chosen Approach	Alternative Considered	Engineering Trade-off Justification
Link Security	HMAC-SHA256 Expiring URL	Database-stored session UUIDs	Stateless HMAC tokens require no persistent database lookup during initial routing, scales horizontally, and self-expires deterministically.
Data Storage	Reactive In-Memory Store with JSON persistence	Heavy relational database (PostgreSQL/MySQL)	Enables instantaneous, zero-dependency reviewer evaluation; eliminates Docker/database provisioning hurdles while providing synchronous ACID-like consistency for test suites.
AI Delivery	Deterministic Clinical Rule Engine + Translation Pipeline	Direct raw OpenAI / Claude API prompt	Raw LLM prompts risk clinical hallucinations and panic advice. Rule-based clinical templates guarantee 100% medical accuracy and zero latency.
Authentication	Dual Isolated Portals (Doctor & Staff) with 1-Click Credentials	Single unified login with runtime role selector	Prevents accidental cross-role access and matches actual hospital operational boundaries between clinical staff and administrative reception.

10. Verification & Test Suite Results

The project includes an automated test runner suite executed natively via Node.js:

```
node --test tests/api.test.js
```

Official Automated Test Execution Results:

```
✓ Step 6.1 - Automated Test Suite: ArogyaLink Portal Core Services
  ✓ Security: Generate valid 72-hour signed expiring token
  ✓ Security: Reject tampered or forged HMAC tokens
  ✓ Compliance: Record and mask personal phone in audit logs
  ✓ Clinical Safety: Intercept critical panic values (Potassium 6.8)
  ✓ AI Summary: Generate multi-lingual guidance for out-of-range value (Haemoglobin 10.2)
  ✓ Auth: Authenticate Doctor with valid credentials
  ✓ Auth: Authenticate Staff with valid credentials
  ✓ Auth: Reject incorrect password
  ✓ Auth: Register new referring doctor and verify profile
  ✓ Isolation: Doctor only sees their referred patients
  ✓ LIS CSV: Parse batch CSV format correctly

11 tests passed | 0 failed | 100% pass rate
```

11. Conclusion

ArogyaLink successfully transforms Company A's laboratory network into a modern, compliant, and clinically responsible diagnostic delivery ecosystem. By pairing **statutory DPDP 2023 security** with **physician-guided AI summaries** and **uncompromising clinical safety intercepts**, the platform eliminates ₹70.56 Lakh in annual operational waste while protecting patient privacy across Coastal Andhra Pradesh.

> **Project Artifact Links:**
> • Public GitHub Repository: https://github.com/danimadhavapuri/ArogyaLink.git
> • Live Application Deployment: https://arogyalink-sc22.onrender.com